



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2024 Year 11 Course Assessment Task 2

Friday, 28th June 2024

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Answer each question in a separate writing booklet. Extra writing booklets are available.
- Write using a blue or a black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- A NESA Reference Sheet is provided.

Class (please ✓)

- ☐ 11MAX.1 – Miss J. Kim
- ☐ 11MAX.2 – Miss Chaussivert
- ☐ 11MAX.3 – Miss Lee
- ☐ 11MAX.4 – Mr Tran
- ☐ 11MAX.5 – Ms Ham
- ☐ 11MAX.6 – Mr Ho

NESA STUDENT #

Marker's use only.

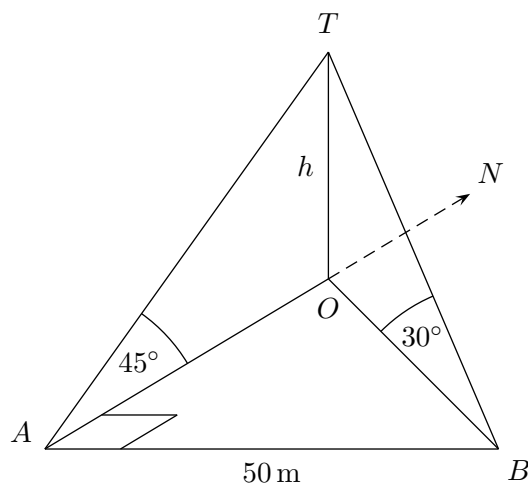
QUESTION	1	2	3	4	5	6	7	Total	%
MARKS	$\overline{6}$	$\overline{9}$	$\overline{3}$	$\overline{8}$	$\overline{6}$	$\overline{3}$	$\overline{12}$	$\overline{47}$	

Question 1 (6 Marks)

Commence a NEW writing booklet

Marks

A person is standing at point A , which is due south of a tower OT that is h metres tall. The person walks 50 metres east to point B . The angle of elevation at points A and B to the top of the tower are 45° and 30° respectively.



(a) Express OA and OB in terms of h . **2**

(b) Show that $h = 25\sqrt{2}$ metres. **2**

(c) Find the bearing of B from the base of the tower O , correct to the nearest minute. **2**

Question 2 (9 Marks)

Commence a NEW writing booklet

Marks

(a) Differentiate the following with respect to x :

i. $x^2\sqrt{x}$ **2**

ii. $(2x^2 + 1)^6$ **2**

iii. $\frac{1 + 7x}{x^3}$ **2**

(b) If $f(x) = \frac{1}{x-1}$, show that $f'(x) = -\frac{1}{(x-1)^2}$ by differentiation from first principles. **3**

Question 3	(3 Marks)	Commence a NEW writing booklet	Marks
Solve for x :			3
	$2 \log_3(x) = \log_3(6 - x)$		

Question 4	(8 Marks)	Commence a NEW writing booklet	Marks
(a) For $f(x) = x^2 - 6x + 11$			
i. Find the range of $f(x)$.			1
ii. State the largest domain of $f(x)$ containing $x = 4$ such that it is monotonic increasing.			1
iii. Find an expression for the inverse function, $f^{-1}(x)$ by using the restricted domain from part (ii).			3
(b) Find the Cartesian equation of the curve defined by the parametric equations:			3

$$\begin{cases} x = 3 \sin \theta - 2 \\ y = 4 \cos \theta + 1 \end{cases}$$

Question 5	(6 Marks)	Commence a NEW writing booklet	Marks
Consider the curve			
	$f(x) = -3x^5 + 5x^3 + 1$		
(a) Find the coordinates of the stationary points and determine their nature.			4
(b) Hence sketch the graph of $y = f(x)$, showing the stationary points, and the y -intercept. (Do NOT find the x -intercept(s).)			2

Examination continues overleaf...

Question 6 (3 Marks) Commence a NEW writing booklet **Marks**

Prove $(1 + \sin \theta + \cos \theta)^2 = 2(1 + \sin \theta)(1 + \cos \theta)$. **3**

Question 7 (12 Marks) Commence a NEW writing booklet **Marks**

(a) Given the vectors $\underline{a} = 2\underline{i} + \underline{j}$, $\underline{b} = 4\underline{i} - 2\underline{j}$ and $\underline{c} = \underline{a} - \underline{b}$.

i. Find the value of $|\underline{c}|$. **1**

ii. Hence, find a vector in the same direction as $\underline{c}$ that has a magnitude of 3 units. **2**

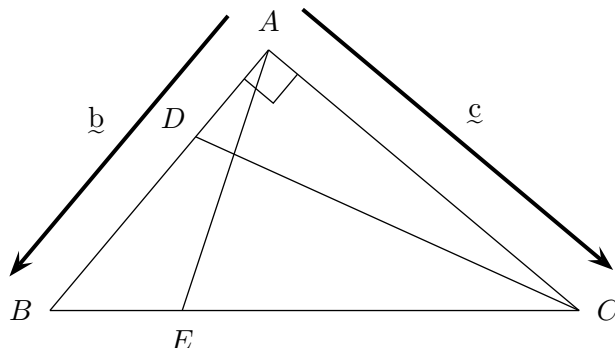
(b) The projection of $\underline{u} = \begin{pmatrix} 3 \\ \lambda \end{pmatrix}$ onto $\underline{v} = \begin{pmatrix} 8 \\ -15 \end{pmatrix}$ has length $\frac{159}{17}$.

i. Show that $|24 - 15\lambda| = 159$. **2**

ii. It is given that $A(3, \lambda)$ and $B(8, -15)$. Hence, find the shortest distance from $A(3, \lambda)$ to the line OB where $\lambda < 0$. **3**

(c) In the right-angled triangle ABC , point D divides the interval AB so that $AD : DB = 1 : 2$, and point E divides the interval BC so that $BE : EC = 1 : 3$. **4**

Let $\underline{b} = \overrightarrow{AB}$ and $\underline{c} = \overrightarrow{AC}$.



Show that $\overrightarrow{AE} \cdot \overrightarrow{CD} = \frac{1}{4} \left(|\overrightarrow{AB}|^2 - |\overrightarrow{AC}|^2 \right)$.

End of paper.

Question 1 (6 marks) (LEE)

(a) (2 marks)

✓ [1] for finding OA

✓ [1] for finding OB

$$\begin{aligned}\tan 45^\circ &= \frac{h}{OA} \\ OA &= \frac{h}{\tan 45^\circ} \\ &= h \quad \checkmark\end{aligned}$$

$$\begin{aligned}\tan 30^\circ &= \frac{h}{OB} \\ \frac{1}{\sqrt{3}} &= \frac{h}{OB} \\ OB &= \sqrt{3}h \quad \checkmark\end{aligned}$$

(b) (2 marks)

✓ [1] for applying Pythagoras' Theorem

✓ [1] for final answer

$$\begin{aligned}OA^2 + 50^2 &= OB^2 \\ h^2 + 2500 &= (\sqrt{3}h)^2 \quad \checkmark \\ 2h^2 &= 2500 \\ h &= \pm\sqrt{1250} \\ &= 25\sqrt{2} \quad (h > 0) \quad \checkmark\end{aligned}$$

(c) (2 marks)

✓ [1] for finding $\angle AOB$

✓ [1] for finding the bearing, correct to the nearest minute

$$\begin{aligned}\tan \angle AOB &= \frac{50}{OA} \\ &= \frac{50}{25\sqrt{2}} \\ \angle AOB &= \tan^{-1}\left(\frac{2}{\sqrt{2}}\right) \quad \checkmark \\ \text{Bearing of B} &= 180^\circ - \angle AOB \\ &= 180^\circ - \tan^{-1}\left(\frac{2}{\sqrt{2}}\right) \\ &= 125^\circ 16' \quad \checkmark\end{aligned}$$

Question 2 (9 marks) (Ham)

(a) i. (2 marks)

✓ [1] for applying the appropriate differentiation rule

✓ [1] for final answer

$$\begin{aligned}
 \frac{d}{dx}(x^2\sqrt{x}) &= \frac{d}{dx}(x^{\frac{5}{2}}) \\
 &= \frac{5}{2}x^{\frac{3}{2}} \quad \checkmark \\
 &= \frac{5}{2}x\sqrt{x} \quad \checkmark
 \end{aligned}$$

ii. (2 marks)

✓ [1] for applying the chain rule

✓ [1] for final answer

$$\begin{aligned}
 \frac{d}{dx}((2x^2 + 1)^6) &= 6(2x^2 + 1)^5(4x) \quad \checkmark \\
 &= 24x(2x^2 + 1)^5 \quad \checkmark
 \end{aligned}$$

iii. (2 marks)

✓ [1] for applying appropriate differentiation rule

✓ [1] for final answer

$$\begin{aligned}
 \frac{d}{dx}\left(\frac{1+7x}{x^3}\right) &= \frac{d}{dx}(x^{-3} + 7x^{-2}) \\
 &= -3x^{-4} + 7(-2)x^{-3} \quad \checkmark \\
 &= -\frac{3}{x^4} - \frac{14}{x^3} \quad \checkmark
 \end{aligned}$$

(b) (3 marks)

✓ [1] for applying differentiation by first principles

✓ [1] for obtaining $\frac{-1}{(x-1)(x+h-1)}$

✓ [1] for final answer

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h-1} - \frac{1}{x-1}}{h} \quad \checkmark \\
 &= \lim_{h \rightarrow 0} \frac{\frac{(x-1) - (x+h-1)}{(x-1)(x+h-1)}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{-h}{h(x-1)(x+h-1)} \\
 &= \lim_{h \rightarrow 0} -\frac{1}{(x-1)(x+h-1)} \quad \checkmark \\
 &= -\frac{1}{(x-1)^2} \quad \checkmark
 \end{aligned}$$

Question 3 (3 marks) (Lam)

- $\checkmark$ [1] for applying logarithm laws to obtain quadratic equation
 $\checkmark$ [1] for factorising the quadratic equation
 $\checkmark$ [1] for final answer

$$\begin{aligned}
 2\log_3(x) &= \log_3(6-x) \\
 \log_3(x^2) &= \log_3(6-x) \\
 x^2 &= 6-x \quad \checkmark \\
 x^2 + x - 6 &= 0 \\
 (x+3)(x-2) &= 0 \quad \checkmark \\
 x &= -3 \text{ or } 2 \\
 \text{Since } 0 < x < 6 : x &= 2 \quad \checkmark
 \end{aligned}$$

Question 4 (8 marks) (Chaussivert)

- (a) i. (1 mark)
 $\checkmark$ [1] for final answer

$$f(x) \geq 2 \text{ or } f(x) \in [2, \infty) \quad \checkmark$$

- ii. (1 mark)
 $\checkmark$ [1] for final answer

$$x \geq 3 \text{ or } x \in [3, \infty) \quad \checkmark$$

iii. (3 marks)

- ✓ [1] for using difference of 2 squares
- ✓ [1] for switching the positions of x and y
- ✓ [1] for final answer

$$y = x^2 - 6x + 11$$

$$= (x - 3)^2 + 2 \quad \checkmark$$

$$x = (y - 3)^2 + 2 \quad \checkmark$$

$$y - 3 = \pm\sqrt{x - 2}$$

$$y = 3 \pm \sqrt{x - 2}$$

Since $D_{f(x)} : x \geq 3$

Then, $R_{f^{-1}(x)} : y \geq 3$

$$y = 3 + \sqrt{x - 2} \quad \checkmark$$

(b) (3 mark)

- ✓ [1] for making $\sin \theta$ the subject
- ✓ [1] for making $\cos \theta$ the subject
- ✓ [1] for final answer

$$\sin \theta = \frac{x + 2}{3} \quad \checkmark$$

$$\cos \theta = \frac{y - 1}{4} \quad \checkmark$$

Since $\sin^2 \theta + \cos^2 \theta = 1$

$$\left(\frac{x + 2}{3}\right)^2 + \left(\frac{y - 1}{4}\right)^2 = 1 \quad \checkmark$$

Question 5 (6 marks) (J.Kim)

(a) (4 marks)

- ✓ [1] for correct derivative
- ✓ [1] for finding stationary points
- ✓ [1] for nature of stationary points
- ✓ [1] for coordinates with correct nature

$$\begin{aligned}
 f'(x) &= -15x^4 + 15x^2 && \checkmark \\
 &= 15x^2(-x^2 + 1) \\
 &= 15x^2(1 - x)(1 + x)
 \end{aligned}$$

Stationary Points occur when $f'(x) = 0$:

$$x = 0, \quad f'(0) = 1 \Rightarrow (0, 1)$$

$$x = 1, \quad f'(1) = 3 \Rightarrow (1, 3)$$

$$x = -1, \quad f'(-1) = -1 \Rightarrow (-1, -1) \quad \checkmark$$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
$f'(x)$	-180	0	2.81	0	2.81	0	-180
$f(x)$	$\searrow$	-	$\nearrow$	-	$\nearrow$	-	$\searrow$

✓

 $\therefore (0, 1)$ is a HPOI $(1, 3)$ is a local max $(-1, -1)$ is a local min ✓

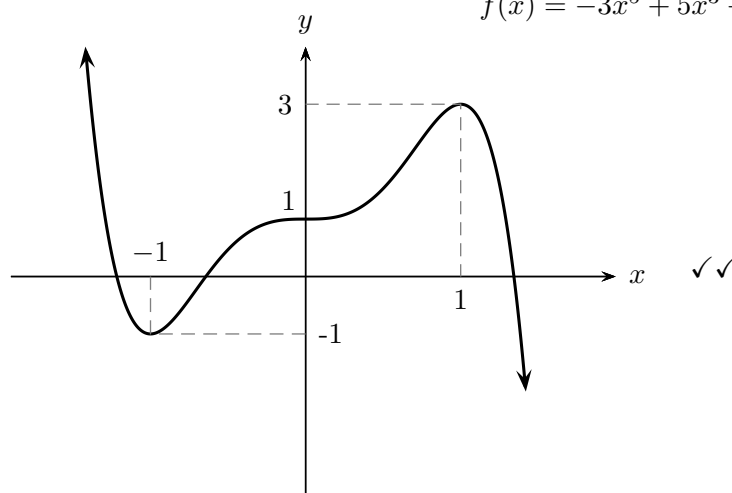
(b) (2 marks)

- ✓ [1] for shape and y -intercept
- ✓ [1] for stationary points

$$f(0) = 1$$

 $\therefore y$ - intercept at $(0, 1)$

$$f(x) = -3x^5 + 5x^3 + 1$$



Question 6 (3 marks) (J.Kim)

- ✓[1] for correct expansion
 ✓[1] for correct substitution
 ✓[1] for correct factorisation to get RHS

$$\begin{aligned}
 \text{LHS} &= (1 + \sin \theta + \cos \theta)^2 \\
 &= 1 + (\sin \theta + \cos \theta)^2 + 2(\sin \theta + \cos \theta) \quad \checkmark \\
 &= 1 + (\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta) + 2 \sin \theta + 2 \cos \theta \\
 &= 2 + 2 \sin \theta \cos \theta + 2 \sin \theta + 2 \cos \theta \quad \checkmark \\
 &= 2(1 + \sin \theta \cos \theta + \sin \theta + \cos \theta) \\
 &= 2[1 + \sin \theta + \cos \theta(1 + \sin \theta)] \quad \checkmark \\
 &= 2(1 + \sin \theta)(1 + \cos \theta) \\
 &= \text{RHS}
 \end{aligned}$$

Question 7 (12 marks) (Ho and Tran)

- (a) i. (1 mark) (Ho)
 ✓ [1] for correct answer

$$\begin{aligned}
 |\underline{\mathbf{c}}| &= |-2\underline{\mathbf{i}} + 3\underline{\mathbf{j}}| \\
 &= \sqrt{(-2)^2 + 3^2} \\
 &= \sqrt{13} \quad \checkmark
 \end{aligned}$$

- ii. (2 marks) (Ho)
 ✓ [1] for finding a unit vector in the same direction as $\underline{\mathbf{c}}$
 ✓ [1] for final answer

$$\begin{aligned}
 \hat{\underline{\mathbf{c}}} &= \frac{1}{\sqrt{13}} (-2\underline{\mathbf{i}} + 3\underline{\mathbf{j}}) \quad \checkmark \\
 3\hat{\underline{\mathbf{c}}} &= \frac{3}{\sqrt{13}} (-2\underline{\mathbf{i}} + 3\underline{\mathbf{j}}) \quad \checkmark
 \end{aligned}$$

- (b) i. (2 marks) (Ho)
- ✓ [1] for finding the $\text{proj}_{\underline{v}} \underline{u}$
 - ✓ [1] for final answer

$$\begin{aligned}\text{proj}_{\underline{v}} \underline{u} &= \frac{\begin{pmatrix} 3 \\ \lambda \end{pmatrix} \cdot \begin{pmatrix} 8 \\ -15 \end{pmatrix}}{\left(\sqrt{8^2 + (-15)^2}\right)^2} \begin{pmatrix} 8 \\ -15 \end{pmatrix} \\ &= \frac{24 - 15\lambda}{289} \begin{pmatrix} 8 \\ -15 \end{pmatrix} \quad \checkmark \\ \left|\text{proj}_{\underline{v}} \underline{u}\right| &= \frac{|24 - 15\lambda|}{289} \left|\begin{pmatrix} 8 \\ -15 \end{pmatrix}\right| \\ \frac{|24 - 15\lambda|}{289} \sqrt{289} &= \frac{159}{17} \\ \frac{|24 - 15\lambda|}{17} &= \frac{159}{17} \\ |24 - 15\lambda| &= 159 \quad \checkmark\end{aligned}$$

- ii. (3 marks) (Ho)
- ✓ [1] for finding the value of λ
 - ✓ [1] for a correct equation (or vector equation) to find
 - ✓ [1] for final answer

From part (i) : $24 - 15\lambda = \pm 159$

$$\begin{aligned}\lambda &= \frac{24 \pm 159}{15} \\ &= -9 \text{ or } \frac{61}{5} \\ &= -9 \quad (\lambda < 0) \quad \checkmark \\ d_{OA} &= \sqrt{3^2 + (-9)^2} \\ &= \sqrt{90} \\ d_{AB}^2 &= d_{OA}^2 - \left(\left|\text{proj}_{\underline{v}} \underline{u}\right|\right)^2 \quad \checkmark \\ &= (\sqrt{90})^2 - \left(\frac{159}{17}\right)^2 \\ &= \frac{729}{289} \\ d_{AB} &= \pm \frac{27}{17} \\ &= \frac{27}{17} \text{ units } (d > 0) \quad \checkmark\end{aligned}$$

(c) (4 marks) (Tran)

- ✓ [1] for finding $\overrightarrow{AE}$
- ✓ [1] for finding $\overrightarrow{CD}$
- ✓ [1] for using $\mathbf{b} \cdot \mathbf{c} = 0$
- ✓ [1] for final answer

$$\overrightarrow{BC} = \mathbf{c} - \mathbf{b}$$

$$\begin{aligned}\overrightarrow{AE} &= \mathbf{b} + \frac{1}{4}(\mathbf{c} - \mathbf{b}) \\ &= \frac{1}{4}(\mathbf{c} + 3\mathbf{b}) \quad \checkmark\end{aligned}$$

$$\begin{aligned}\overrightarrow{CD} &= \frac{1}{3}\mathbf{b} - \mathbf{c} \\ &= \frac{1}{3}(\mathbf{b} - 3\mathbf{c}) \quad \checkmark\end{aligned}$$

$$\begin{aligned}\overrightarrow{AE} \cdot \overrightarrow{CD} &= \frac{1}{12}(\mathbf{c} + 3\mathbf{b}) \cdot (\mathbf{b} - 3\mathbf{c}) \\ &= \frac{1}{12}(3|\mathbf{b}|^2 - 3|\mathbf{c}|^2 - 8\mathbf{b} \cdot \mathbf{c})\end{aligned}$$

$$\text{Since } \mathbf{b} \cdot \mathbf{c} = 0 \quad \checkmark$$

$$\begin{aligned}\overrightarrow{AE} \cdot \overrightarrow{CD} &= \frac{1}{4}(|\mathbf{b}|^2 - |\mathbf{c}|^2) \\ &= \frac{1}{4}(|\overrightarrow{AB}|^2 - |\overrightarrow{AC}|^2) \quad \checkmark\end{aligned}$$